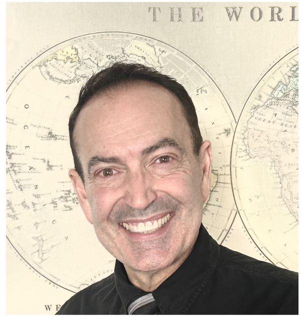
PETER ALEXANDER



Peter Alexander is a mathematics educator, curriculum designer, and quantitative consultant with experience spanning advanced secondary, collegiate, and interdisciplinary instruction. He completed his M.A. in Mathematics at UC Berkeley and completed all coursework and comprehensive qualifying examinations required for the Ph.D. in Mathematics. His teaching emphasizes conceptual understanding, mathematical communication, and the development of deep problem-solving intuition rather than procedural memorization alone.

Peter is especially interested in helping students experience mathematics as a language for representing structures, patterns, transformations, and real-world systems. His instructional work frequently integrates data science, computational modeling, quantitative finance, cryptography, machine learning, and algorithmic thinking in ways that make advanced concepts accessible and intellectually engaging. In addition to traditional instruction, he has developed original curriculum materials, interactive assessment systems, and computational learning tools designed to strengthen conceptual engagement and student ownership of learning.

Over the course of his career, Peter has mentored students across a wide range of subjects and accelerated pathways while helping them build confidence, independence, and long-term intellectual maturity.

SUBJECTS & AREAS COVERED

Pre-Algebra • Algebra I & II • Geometry • Trigonometry • Precalculus • AP Calculus AB/BC • Multivariable Calculus • Linear Algebra • Differential Equations • AP Statistics • Discrete Mathematics • Number Theory • Quantitative Finance • Data Science • AP Computer Science A • Python Programming • Machine Learning • AP Physics 1 & 2 • AP Physics C

TESTS COVERED

SAT • ACT • GRE • GMAT • AMC • AIME • AP Exams • CBEST

SPECIALIZATIONS

Inquiry-Based Mathematics Instruction • Advanced & Gifted STEM Mentorship • Computational Thinking • Quantitative Modeling • Mathematical Programming • Standardized Test Strategy • Interactive Assessment Design • Phenomenological Representation and Abstraction